

Tuesday 12:02 AM

**Urban Flood
Nowcasting
System
(Drainage and
Rainfall
Coupling)**

Description	Urban flooding in major Indian metros like Mumbai, Delhi, and Chennai has become an annual crisis. Traditional Numerical Weather Prediction (NWP) models fall short because knowing how much rain will fall does not automatically translate into knowing where the streets will flood. Urban flooding is a hyper-local phenomenon dictated by micro-topography, concrete imperviousness, and heavily strained, invisible drainage networks. Currently, municipal bodies lack real-time, street-level predictive systems. Consequently, cities are caught off guard by rapid water accumulation, leading to severe traffic gridlocks, economic disruption, and loss of life. The challenge is to design a high-resolution, real-time Urban Flood Nowcasting System (0–3 hour lead time) capable of predicting street-level inundation before it happens. Participants must move away from isolated weather models and instead build a coupled framework. This system must fuse real-time rainfall nowcasts with high-resolution Digital Elevation Models (DEM) and a graph-based mathematical model of the city’s underground drainage network. By mapping how water flows, accumulates, and surcharges across concrete surfaces and drainage nodes, the solution should pinpoint exactly which streets or intersections will flood. Develop a pipeline that takes high-resolution rainfall nowcasts (from Doppler Weather Radars) and instantly routes that volume across a 2D surface terrain model. Represent the city’s stormwater drain network as a directed graph (nodes as manholes/inlets, edges as pipes/canals). The model must calculate hydraulic capacity and predict where blockages or overcapacity will cause backflow onto the streets. A dynamic, web-based GIS dashboard showing real-time, street-by-street flooding projections (e.g., water depth estimations in centimeters) with a 0–3 hour forward-looking window. An API utility that can interface with navigation maps to suggest flood-safe alternative routes for emergency services, public transit, and commuters during heavy downpours.
Organization	Ministry of Earth Sciences (MoES)
Department	National Centre for Medium Range Weather Forecasting (NCMRWF)
Category	Software
Theme	Disaster Management

explain me this problem statement
deply, i do not want answer/solution, only problem statement explain, if
possible multiple parts me break kr do jisse aur acha explain kro

Bilkul. Is problem statement ko **solution soche bina**, sirf “**problem actually hai kya?**” ke perspective se samajhte hain.

Main ise parts mein tod raha hoon, aur technical words ko bhi simple language mein explain karunga.

1. Sabse pehle: Problem ek line mein kya hai?

Problem basically ye keh raha hai:

Agar kisi city mein agle 0–3 ghante mein bahut baarish hone wali hai, toh pehle se predict karo ki city ki kaunsi exact street/intersection par kitna paani jama hoga aur flooding hogi.

Important point:

Sirf baarish predict nahi karni hai.

Example:

Weather department bolta hai:
“Mumbai mein next 1 hour mein 50 mm rain hogi.”

Ye useful hai, **but enough nahi hai.**

Municipality ko actually pata hona chahiye:

“50 mm rain ke baad Andheri ki Street A par 15 cm water hoga, Street B par 40 cm, aur Intersection C completely flood ho jayega.”

Ye difference hi poore problem statement ka core hai.

2. Current problem kya hai?

Problem statement ke according, Indian metro cities jaise:

- Mumbai
- Delhi
- Chennai

mein urban flooding baar-baar hoti hai.







7

Heavy rain ke baad:

- roads par paani bhar jata hai
- traffic jam ho jata hai
- vehicles stuck ho jati hain
- public transport affect hota hai
- businesses/economy affect hoti hai
- emergency services ko problem hoti hai
- extreme cases mein lives bhi risk mein hoti hain

Lekin problem sirf "bahut rain" nahi hai.

Suppose:

Road A:

Rain = 50 mm

Drainage = excellent

Terrain = slope hai

→ Paani quickly drain ho sakta hai.

Road B:

Rain = 50 mm

Drainage = already overloaded

Terrain = low-lying
Surface = mostly concrete

→ Yahan severe flooding ho sakti hai.

Dono jagah **same rainfall** hai, lekin flooding completely different hai.

3. Traditional weather forecasting se problem solve kyun nahi hoti?

Statement mein ek important line hai:

“Knowing how much rain will fall does not automatically translate into knowing where the streets will flood.”

Isko bahut simply samjho.

Weather model answer karta hai:

"Rain kitni hogi?"

Flooding system ko answer karna hai:

"Us rain ke baad paani exactly kahan jayega?"

Ye do alag problems hain.

Example

Maan lo:

Rainfall = 80 mm

Ab ye information alone nahi batati:

Street A → 5 cm water

Street B → 20 cm water

Street C → 50 cm water

Street D → no flooding

Kyun?

Because water ka behaviour depend karta hai:

- land ki height
- road ka slope

- concrete kitna hai
 - soil kitna water absorb kar sakti hai
 - drains ki capacity
 - drains blocked hain ya nahi
 - water kis direction mein flow karega
 - drainage network mein downstream blockage hai ya nahi
-

4. "Hyper-local phenomenon" ka kya meaning hai?

Statement mein bola hai:

Urban flooding is a **hyper-local phenomenon**

Hyper-local = bahut small/local area ke level par different behaviour.

Example:

Ek city mein:

Area A → flood

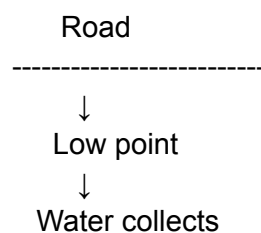
Area B → normal

Area C → severe flood

Area D → slight waterlogging

Even though sabko approximately same rain mili.

Ek hi road ke andar bhi difference ho sakta hai:



Agar road ka ek portion thoda lower hai, toh water wahi accumulate ho sakta hai.

Isliye problem **city-level flood prediction** nahi maang rahi.

Ye **street-level prediction** maang rahi hai.

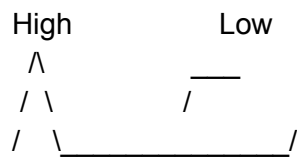
5. "Micro-topography" kya hai?

Ye ek important term hai.

Topography

Land ki shape/height.

For example:

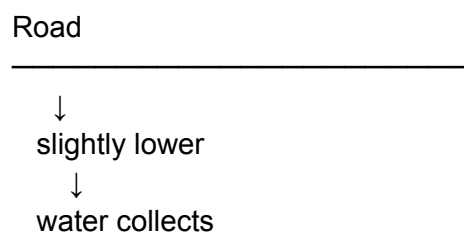


Water generally lower areas ki taraf move karega.

Micro-topography

Bahut chhote area ki detailed height differences.

For example:



Ek road ka 10–20 cm elevation difference bhi water accumulation ko affect kar sakta hai.

Problem statement basically keh raha hai:

Flooding predict karne ke liye city ki detailed surface shape samajhni padegi.

6. "Imperviousness" kya hai?

Ye bhi important concept hai.

Impervious surface = jahan water easily ground mein absorb nahi hota.

Examples:

- concrete roads

- buildings
- parking lots
- asphalt
- paved areas

Compare:

Open soil

Rain



Ground absorbs some water

Concrete

Rain



→→→ water flows away

Urban areas mein concrete bahut zyada hota hai.

Isliye rainwater ka large portion directly surface runoff ban sakta hai.

7. "Strained drainage networks" ka kya matlab hai?

City ke neeche stormwater drainage system hota hai.

Simple example:

Rain



Road



Drain inlet



Pipe



Bigger pipe



Main drainage channel



River/sea

Normal rain mein system handle kar leta hai.

Lekin heavy rainfall mein:

Incoming water



Drain capacity



MAX



Extra water



Street flooding

Yaani:

water aa raha hai > drainage system jitna water handle kar sakta hai

toh water surface par accumulate hone lagta hai.

8. "Invisible drainage networks" kyun bola?

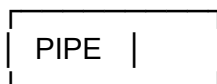
Drainage system mostly underground hota hai.

Road ke neeche:

ROAD



inlet



underground

Normal person ko ye network visible nahi hota.

Lekin flooding ke liye ye extremely important hai.

Problem statement therefore keh raha hai:

Surface ko bhi samjho + underground drainage ko bhi samjho.

9. Actual challenge kya hai?

Ab statement ka main challenge aata hai.

System ko **0–3 hour lead time** ke saath flooding predict karni hai.

Lead time

Lead time = event hone se pehle available warning time.

Example:

Current time:

10:00 AM

System predict karta hai:

10:30 AM → Street A flooding

11:00 AM → Street B flooding

12:00 PM → Intersection C severe flooding

Toh system ke paas **0–3 hours ka forward-looking window** hai.

10. "Nowcasting" ka meaning

Nowcasting basically:

Very near future ka prediction.

Normal weather forecasting:

Tomorrow's weather

Next 7 days

Nowcasting:

Next few minutes

Next 1 hour

Next 2 hours
Next 3 hours

Yahan specifically:

Urban flood nowcasting = next 0–3 hours mein flooding kahan hogi uska prediction.

11. Problem statement system se kya input expect kar raha hai?

Statement specifically kuch important data sources mention karta hai.

Sabse pehla:

High-resolution rainfall nowcasts

Yaani system ko information milegi:

Next few hours mein different locations par rainfall kitni expected hai.

Example:

Area A → 40 mm
Area B → 70 mm
Area C → 90 mm
Area D → 20 mm

Aur ye rainfall information **Doppler Weather Radar** se aa sakti hai.

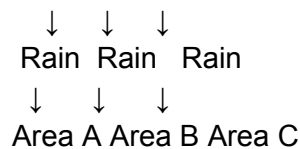
12. Doppler Weather Radar kya hai?

Simple terms mein:

Weather radar atmosphere ko observe karta hai aur rainfall ke location/intensity/movement ke baare mein information provide kar sakta hai.

Conceptually:

RADAR
|



Problem statement chahta hai ki system **real-time rainfall nowcast** ko use kare.

13. Second important input: DEM

Statement mein hai:

high-resolution Digital Elevation Models (DEM)

DEM = Digital Elevation Model

Basically:

Area ke different points ki elevation/height ka digital representation.

Imagine city ko grid mein divide kar diya:

100m	101m	102m	103m
98m	99m	100m	101m
95m	96m	98m	99m
92m	93m	95m	97m

Ab system ko idea milta hai:

Water naturally kis direction mein move karega?

Because water generally higher elevation se lower elevation ki taraf move karta hai.

14. Third important thing: Drainage network

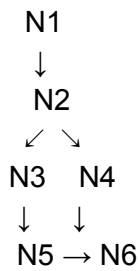
Problem statement drainage network ko **graph** ke form mein represent karne ki baat karta hai.

Yahan graph ka matlab:

Node → drainage point/manhole/inlet

Edge → pipe/canal connecting two points

Example:



Yahan:

- N1, N2, N3... = drainage nodes
- arrows/lines = pipes/canals

Problem statement specifically bolta hai:

directed graph

Yaani water flow ki direction important hai.

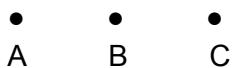
15. Node kya ho sakta hai?

Statement ke according node examples:

- manholes
- inlets

Imagine road par:

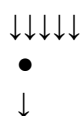
Road



Ye drainage entry points ho sakte hain.

Rainwater:

Road



pipe

16. Edge kya hai?

Do drainage nodes ke beech connection:

Node A —————→ Node B

Ye represent kar sakta hai:

- underground pipe
- drain
- canal

Aur har connection ki capacity ho sakti hai.

Conceptually:

A —[pipe capacity]——→ B

17. "Hydraulic capacity" ka kya meaning hai?

Simple language:

Drain/pipe maximum kitna water safely carry kar sakta hai.

Example:

Pipe capacity = 100 units/hour

Agar incoming water:

80 units/hour

→ manageable.

But:

150 units/hour

→ pipe overloaded.

So:

Incoming water > Drain capacity

↓
Overcapacity

↓
Water backs up

↓
Street flooding

18. "Blockage" kya hai?

Suppose drainage pipe normally:

A —————→ B

Water easily flow karta hai.

But blockage:

A ————X————→ B

Ab water आगे nahi ja pa raha.

Result:

Water entering

↓
↓

BLOCKAGE X

↓

water accumulates

↓

backflow

↓

street flooding

Problem statement chahta hai ki system **blockages ya overcapacity ke impact** ko consider kare.

19. "Surcharge" / "Backflow" ka meaning

Ye terminology important hai.

Normally:

Street
↓
Drain
↓
Pipe
↓
Outlet

Water street se drain mein ja raha hai.

But agar downstream system overloaded hai:

Street
↓
Drain
↓
Pipe
↓
XXXX overloaded

Water aage nahi ja pa raha.

Pressure/build-up ki wajah se water drainage system mein reverse direction mein aa sakta hai.

Conceptually:

Normal:

Street → Drain → Pipe → Outlet

Overloaded:

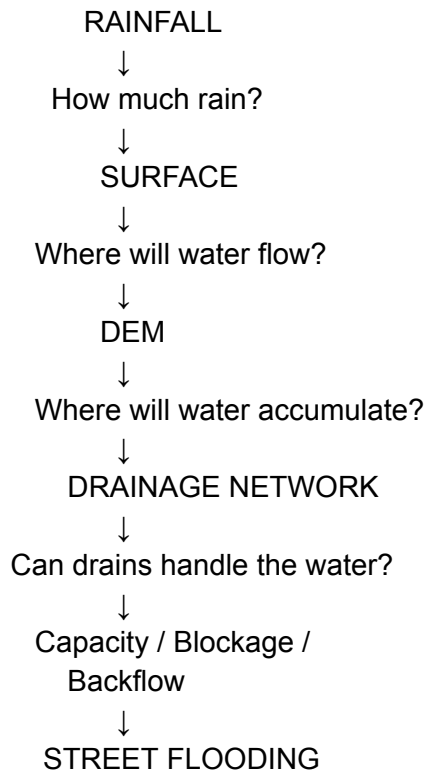
Street ← Drain ← Pipe
↑
water backs up

Isi type ke behaviour ko statement mein **surcharge/backflow** ke context mein mention kiya gaya hai.

20. Sab inputs ko ek saath dekho

Ab poora problem much clearer ho jayega.

System ko roughly ye information samajhni hai:



Yahi problem statement ka heart hai.

21. "2D surface terrain model" kya hai?

Statement kehta hai:

instantly routes that volume across a 2D surface terrain model

Iska matlab city ki surface ko ek detailed 2D representation ki tarah treat karna.

Imagine:

```
+---+---+---+---+
|   |   |   |   |
+---+---+---+---+
|   |   |   |   |
+---+---+---+---+
|   |   |   |   |
+---+---+---+---+
```

Har cell ke paas information ho sakti hai:

- elevation
- land type

- rainfall
- water accumulation

Then conceptually system ko determine karna hai:

Rainfall ka water surface par kis direction mein move karega?

22. Final output kya chahiye?

Ye sabse important part hai.

Problem statement wants:

street-level inundation prediction

Yaani output generic:

✗ "Delhi mein flooding hogi."

It should be much more specific:

Street A → 8 cm

Street B → 23 cm

Street C → 41 cm

Intersection D → 55 cm

Exact numbers sirf example hain — statement khud koi fixed accuracy/value nahi de raha.

23. Water depth estimation

Statement explicitly example deta hai:

water depth estimations in centimeters

So system ideally bata sake:

Location	Predicted depth
----------	-----------------

Street A	5 cm
----------	------

Street B	18 cm
----------	-------

Street C	37 cm
----------	-------

Intersection D	52 cm
----------------	-------

And this prediction future ke liye ho:

+30 min
+60 min
+90 min
+120 min
+180 min

24. GIS dashboard kya hai?

Statement ke end mein:

dynamic, web-based GIS dashboard

GIS

Geographic Information System

Simple language:

Map ke upar geographical information display karna.

Normal map:

Road _____
Road _____

GIS flood dashboard:

CITY MAP

Road A _____


Road B _____


Road C _____


Different areas ko flood severity ke according visually show kiya ja sakta hai.

25. Dashboard "real-time" kyun hona chahiye?

Because rainfall continuously change hoti hai.

Example:

10:00 AM

Rainfall = moderate

Prediction = low flooding

10:30 AM

Rainfall increased

Prediction = moderate flooding

11:00 AM

Heavy rainfall

Prediction = severe flooding

So dashboard static report nahi hona chahiye.

Problem statement wants a **dynamic system**.

26. 0–3 hour forward-looking window

Dashboard ko ideally current situation + near future show karna hai.

Conceptually:

NOW



User dekh sake:

"Agle 2 hours mein kaunsi roads flood hone wali hain?"

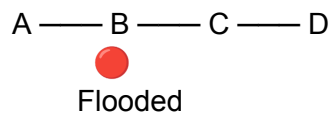
27. Last requirement: Flood-safe routes

Statement ka final part:

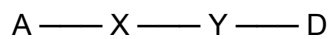
API utility that can interface with navigation maps to suggest flood-safe alternative routes

Iska meaning:

Suppose normal route:



Toh system bol sakta hai:



Yaani:

Flooded road avoid karke alternative route suggest karo.

28. Ye kis ke liye useful hoga?

Statement explicitly mentions:

Emergency services

- ambulance
- fire services
- rescue teams

Unko flooded roads avoid karni hain.

Public transport

Buses etc. ko safer route chahiye.

Commuters

Normal people ko bhi:

"Is road se mat jao, alternate route use karo."

29. "API utility" ka matlab kya hai?

Problem statement complete navigation application banane ko necessarily nahi keh raha.

It says:

API utility that can interface with navigation maps

Meaning system ko aisa interface/API provide karna hai jiske through navigation-related systems flood information consume kar saken.

Conceptually:

Flood Prediction System



API



Navigation / Maps



Safe Route

30. Ab poore problem ko ek story ki tarah samjho

Imagine:

Mumbai mein 7:00 PM hai.

Weather radar predict karta hai:

Next 2 hours heavy rainfall.

Ab traditional information:

"Heavy rain coming."

But municipality ko aur information chahiye.

System ko city ka:

Terrain

pata hai:

Road A → low elevation
Road B → high elevation
Road C → low elevation

System ko:

Drainage network

pata hai:

A → Drain 1 → Drain 2
C → Drain 3 → Drain 2

Aur drainage capacity pata hai.

Ab system rainfall + terrain + drainage ko consider karke future situation determine karta hai.

Conceptually:

Rainfall
+
Terrain
+
Surface characteristics
+
Drainage network
+
Drain capacity/blockage
↓
Future water movement
↓
Flooding locations
↓
Water depth
↓
Map

And finally:

"These streets are likely to flood
within the next 0–3 hours."

31. Problem statement actually kya nahi keh raha?

Ye distinction bahut important hai.

Problem statement **sirf** ye nahi keh raha:

✗ Rainfall prediction system banao.

It is also not simply:

✗ Weather forecasting app banao.

It is not simply:

✗ Flood map banao.

It is asking for a **coupled system**.

Meaning multiple things ko connect karna hai:

Rainfall
+
Terrain
+
Urban surface
+
Drainage network
+
Hydraulic behaviour
↓
Street-level flooding

32. "Coupled framework" ka simple meaning

Coupled = interconnected.

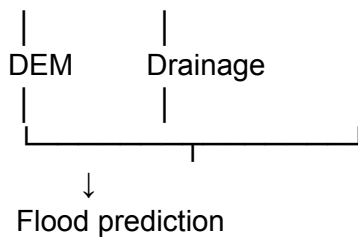
Separate systems:

Rainfall model → rainfall
DEM → elevation
Drainage model → drainage

Problem statement chahta hai ki inko isolated na rakho.

Instead:





Yaani ek component ka information doosre component ke prediction ko influence kare.

33. Ek super-simple analogy

Socho tumhare ghar mein bathtub hai.

Rainfall = **tap se aane wala water**

Drainage = **bathtub ka outlet**

Terrain = **bathtub ka shape/slope**

Blockage = **outlet mein blockage**

Flooding = **bathtub overflow**

Agar:

Water entering < drain capacity

→ no overflow.

Agar:

Water entering > drain capacity

→ water level rises.

Agar outlet blocked hai:

water level rises faster

Urban city basically **bahut huge and complicated bathtub** ki tarah behave kar sakti hai — bas yahan thousands of roads, drains, elevation changes aur rainfall locations hain.

34. Problem statement ke keywords ka translation

Term	Simple meaning
Urban Flood	City mein flooding
Nowcasting	Very near future prediction
0–3 hour lead time	Agle 0–3 hours ki prediction
Hyper-local	Bahut small/local area level
Micro-topography	Chhote area ki detailed height/shape
Imperviousness	Surface jahan water easily absorb nahi hota
Drainage network	Underground drains/pipes ka network
DEM	Land ki elevation ka digital map
Doppler Weather Radar	Rainfall/weather observation system
2D surface model	City surface ka detailed spatial representation
Directed graph	Nodes + directional connections
Node	Manhole/inlet/drainage point
Edge	Pipe/canal connection
Hydraulic capacity	Drain kitna water handle kar sakta hai
Blockage	Drain/pipe mein obstruction
Backflow	Water ka reverse direction mein aana
Surcharge	Drainage system ka overloaded/pressurized state
Inundation	Area ka water se cover hona
Water depth	Road par water ki height
GIS dashboard	Map-based information dashboard
Real-time	Current data ke saath continuously update
Flood-safe route	Flooded area avoid karne wala route

35. Sabse important: Judge actually kya dekhna chahega?

Problem statement ko padhte waqt mentally ye **5 questions** rakho:

Q1. Rain kitni aur kahan ho rahi/expected hai?

→ Rainfall nowcast

Q2. Water surface par kidhar jayega?

→ Terrain / DEM

Q3. Water drain mein kaise jayega?

→ Drainage network

Q4. Drain water handle kar payega ya overflow/backflow hoga?

→ Drainage capacity + blockage/overcapacity

Q5. Ultimately kaunsi street flood hogi aur kitna paani hoga?

→ Street-level inundation prediction

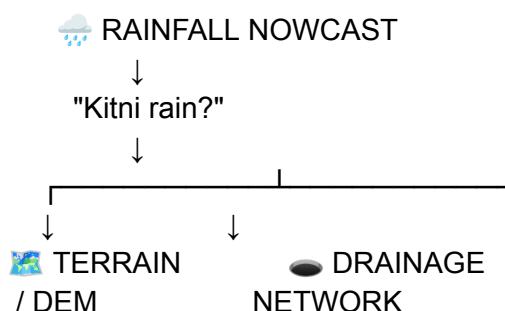
And then:

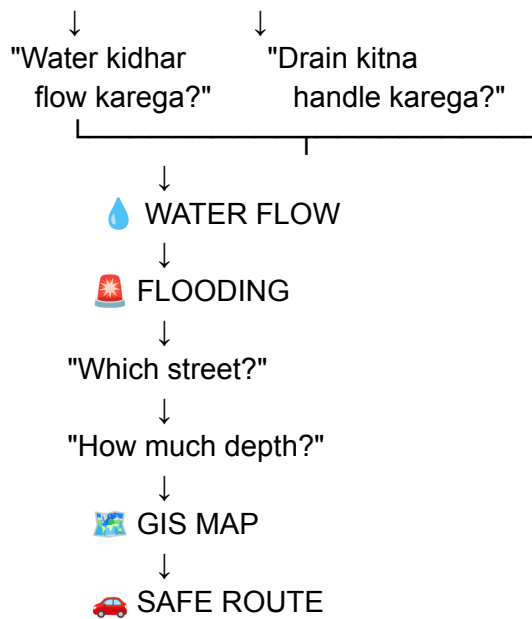
Q6. Logon ko ye information kaise dikhegi/use hogi?

→ GIS dashboard + route API

36. One-line mental model

Agar tumhe poora problem statement **sirf ek diagram** mein yaad rakhna ho:





Basically problem statement ka core ye hai:

Rainfall ko directly flood prediction mat samjho. Rainfall + city ki surface + drainage system ko together samajhkar predict karo ki agle 0–3 hours mein exact streets/intersections par flooding kaise develop hogi.

Aur tumne specifically kaha hai ki **solution nahi chahiye**, isliye maine intentionally ye nahi bataya ki *kaunsa ML model, algorithm, framework, dataset, architecture, ya technology use karni chahiye*. Yeh sirf **problem ko deeply understand karne** ke liye breakdown hai.